

Task 5 – Exploratory Data Analysis (EDA)

Topics Covered

1. Summary Statistics
 - Mean, median, mode, standard deviation.
2. Distributions
 - Histograms and density plots.
3. Correlations
 - Pearson correlation, heatmaps.

Program 1

```
import pandas as pd

data = {'Age': [23, 45, 31, 35, 40],
        'Salary': [25000, 40000, 35000, 30000, 45000]}
df = pd.DataFrame(data)

print(df.describe())
```

Program 2

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.heatmap(df.corr(), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```

☀ Key Takeaways

- Learned how to summarize datasets using statistics.
- Understood distributions through histograms & density plots.
- Explored correlations to find relationships between variables.
- EDA is a crucial step before applying ML models.